

equivalent there, so G has a face f containing f' . By Lemma 4.2.1 (i), however, f lies inside a face f'' of G' . Thus $f' \subseteq f \subseteq f''$ and hence $f' = f = f''$, since both f' and f'' are faces of G' . $\square$

[4.4.1]
[5.1.2]
[7.3.5]

Corollary 4.2.10. *A plane graph with $n \geq 3$ vertices has at most $3n - 6$ edges. Every plane triangulation with n vertices has $3n - 6$ edges.*

Proof. By Proposition 4.2.8 it suffices to prove the second assertion. In a plane triangulation G , every face boundary contains exactly three edges, and every edge lies on the boundary of exactly two faces (Lemma 4.2.2). The bipartite graph on $E(G) \cup F(G)$ with edge set $\{ef \mid e \subseteq G[f]\}$ thus has exactly $2|E(G)| = 3|F(G)|$ edges. According to this identity we may replace ℓ with $2m/3$ in Euler's formula, and obtain $m = 3n - 6$. $\square$

Euler's formula can be useful for showing that certain graphs cannot occur as plane graphs. The graph K^5 , for example, has $10 > 3 \cdot 5 - 6$ edges, more than allowed by Corollary 4.2.10. Similarly, $K_{3,3}$ cannot be a plane graph. For since $K_{3,3}$ is 2-connected but contains no triangle, every face of a plane $K_{3,3}$ would be bounded by a cycle of length ≥ 4 (Proposition 4.2.6). As in the proof of Corollary 4.2.10 this implies $2m \geq 4\ell$, which yields $m \leq 2n - 4$ when substituted in Euler's formula. But $K_{3,3}$ has $9 > 2 \cdot 6 - 4$ edges.

Clearly, along with K^5 and $K_{3,3}$ themselves, their subdivisions cannot occur as plane graphs either:

[4.4.5]
[4.4.6]
[5.1.2]

Corollary 4.2.11. *A plane graph contains neither K^5 nor $K_{3,3}$ as a topological minor.* $\square$

Surprisingly, it turns out that this simple property of plane graphs identifies them among all other graphs: as Section 4.4 will show, an arbitrary graph can be drawn in the plane if and only if it has no (topological) K^5 or $K_{3,3}$ minor.

4.3 Drawings

planar
embedding

drawing

An embedding in the plane, or *planar embedding*, of an (abstract) graph G is an isomorphism between G and a plane graph H . The latter will be called a *drawing* of G . We shall not always distinguish notationally between the vertices and edges of G and of H . In this section we investigate how two planar embeddings of a graph can differ.

How should we measure the likeness of two embeddings $\varrho: G \rightarrow H$ and $\varrho': G \rightarrow H'$ of a planar graph G ? An obvious way to do this is to consider the canonical isomorphism $\sigma := \varrho' \circ \varrho^{-1}$ between H and H' as abstract graphs, and ask how much of their position in the plane

this isomorphism respects or preserves. For example, if σ is induced by a simple rotation of the plane, we would hardly consider ϱ and ϱ' as genuinely different ways of drawing G .

So let us begin by considering any abstract isomorphism $\sigma: V \rightarrow V'$ between two plane graphs $H = (V, E)$ and $H' = (V', E')$, with face sets $F(H) =: F$ and $F(H') =: F'$ say, and try to measure to what degree σ respects or preserves the features of H and H' as plane graphs. In what follows we shall propose three criteria for this in decreasing order of strictness (and increasing order of ease of handling), and then prove that for most graphs these three criteria turn out to agree. In particular, applied to the isomorphism $\sigma = \varrho' \circ \varrho^{-1}$ considered earlier, all three criteria will say that there is essentially only one way to draw a 3-connected graph.

Our first criterion for measuring how well our abstract isomorphism σ preserves the plane features of H and H' is perhaps the most natural one. Intuitively, we would like to call σ ‘topological’ if it is induced by a homeomorphism from the plane $\mathbb{R}^2$ to itself. To avoid having to grant the outer faces of H and H' a special status, however, we take a detour via the homeomorphism $\pi: S^2 \setminus \{(0, 0, 1)\} \rightarrow \mathbb{R}^2$ chosen in Section 4.1: we call σ a *topological isomorphism* between the plane graphs H and H' if there exists a homeomorphism $\varphi: S^2 \rightarrow S^2$ such that $\psi := \pi \circ \varphi \circ \pi^{-1}$ induces σ on $V \cup E$. (More formally: we ask that ψ agree with σ on V , and that it map every plane edge $xy \in H$ onto the plane edge $\sigma(x)\sigma(y) \in H'$. Unless φ fixes the point $(0, 0, 1)$, the map ψ will be undefined at $\pi(\varphi^{-1}(0, 0, 1))$.)

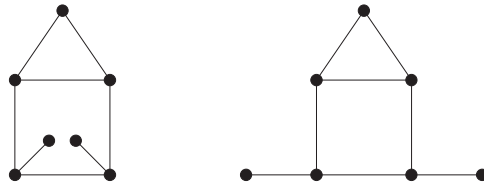


Fig. 4.3.1. Two drawings of a graph that are not topologically isomorphic – why not?

It can be shown that, up to topological isomorphism, inner and outer faces are indeed no longer different: if we choose as φ a rotation of S^2 mapping the π^{-1} -image of a point of some inner face of H to the north pole $(0, 0, 1)$ of S^2 , then ψ maps the rest of this face to the outer face of $\psi(H)$. (To ensure that the edges of $\psi(H)$ are again piecewise linear, however, one may have to adjust φ a little.)

If σ is a topological isomorphism as above, then – except possibly for a pair of missing points where ψ or ψ^{-1} is undefined – ψ maps the faces of H onto those of H' (proof?). In this way, σ extends naturally

to a bijection $\sigma: V \cup E \cup F \rightarrow V' \cup E' \cup F'$ which preserves incidence of vertices, edges and faces.

Let us single out this last property of a topological isomorphism as the second criterion for how well an abstract isomorphism between plane graphs respects their position in the plane: let us call σ a *combinatorial isomorphism* of the plane graphs H and H' if it can be extended to a bijection $\sigma: V \cup E \cup F \rightarrow V' \cup E' \cup F'$ that preserves incidence not only of vertices with edges but also of vertices and edges with faces. (Formally: we require that a vertex or edge $x \in H$ shall lie on the boundary of a face $f \in F$ if and only if $\sigma(x)$ lies on the boundary of the face $\sigma(f)$.)

combinatorial
isomorphism



Fig. 4.3.2. Two drawings of a graph that are combinatorially isomorphic but not topologically – why not?

If σ is a combinatorial isomorphism of the plane graphs H and H' , it maps the face boundaries of H to those of H' . Let us pick out this property as our third criterion, and call σ a *graph-theoretical isomorphism* of the plane graphs H and H' if

graph-
theoretical
isomorphism

$$\{ \sigma(H[f]) : f \in F \} = \{ H'[f'] : f' \in F' \}.$$

Thus, we no longer keep track of *which* face is bounded by a given subgraph: the only information we keep is whether a subgraph bounds some face or not, and we require that σ map the subgraphs that do onto each other. At first glance, this third criterion may appear a little less natural than the previous two. However, it has the practical advantage of being formally weaker and hence easier to verify, and moreover, it will turn out to be equivalent to the other two in most cases.

As we have seen, every topological isomorphism between two plane graphs is also combinatorial, and every combinatorial isomorphism is also graph-theoretical. The following theorem shows that, for most graphs, the converse is true as well:

Theorem 4.3.1.

- (i) Every graph-theoretical isomorphism between two plane graphs is combinatorial. Its extension to a face bijection is unique if and only if the graph is not a cycle.
- (ii) Every combinatorial isomorphism between two 2-connected plane graphs is topological.

Proof. Let $H = (V, E)$ and $H' = (V', E')$ be two plane graphs, put $F(H) =: F$ and $F(H') =: F'$, and let $\sigma: V \rightarrow V'$ be an isomorphism between the underlying abstract graphs. Extend σ to a map $V \cup E \rightarrow V' \cup E'$ by letting $\sigma(xy) := \sigma(x)\sigma(y)$.

(i) If H is a cycle, the assertion follows from the Jordan curve theorem. We now assume that H is not a cycle. Let $\mathcal{B}$ and $\mathcal{B}'$ be the sets of all face boundaries in H and H' , respectively. If σ is a graph-theoretical isomorphism, then the map $B \mapsto \sigma(B)$ is a bijection between $\mathcal{B}$ and $\mathcal{B}'$. By Lemma 4.2.5, the map $f \mapsto H[f]$ is a bijection between F and $\mathcal{B}$, and likewise for F' and $\mathcal{B}'$. The composition of these three bijections is a bijection between F and F' , which we choose as $\sigma: F \rightarrow F'$. By construction, this extension of σ to $V \cup E \cup F$ preserves incidences (and is unique with this property), so σ is indeed a combinatorial isomorphism.

(ii) Let us assume that H is 2-connected, and that σ is a combinatorial isomorphism. We have to construct a homeomorphism $\varphi: S^2 \rightarrow S^2$ which, for every vertex or plane edge $x \in H$, maps $\pi^{-1}(x)$ to $\pi^{-1}(\sigma(x))$. Since σ is a combinatorial isomorphism, $\tilde{\sigma}: \pi^{-1} \circ \sigma \circ \pi$ is an incidence preserving bijection from the vertices, edges and faces⁴ of $\tilde{H} := \pi^{-1}(H)$ to the vertices, edges and faces of $\tilde{H}' := \pi^{-1}(H')$.

$$\begin{array}{ccc}
 S^2 \supseteq \tilde{H} & \xrightarrow{\tilde{\sigma}} & \tilde{H}' \subseteq S^2 \\
 \pi \downarrow & & \downarrow \pi \\
 \mathbb{R}^2 \supseteq H & \xrightarrow{\sigma} & H' \subseteq \mathbb{R}^2
 \end{array}$$

Fig. 4.3.3. Defining $\tilde{\sigma}$ via σ

We construct φ in three steps. Let us first define φ on the vertex set of $\tilde{H}$, setting $\varphi(x) := \tilde{\sigma}(x)$ for all $x \in V(\tilde{H})$. This is trivially a homeomorphism between $V(\tilde{H})$ and $V(\tilde{H}')$.

As the second step, we now extend φ to a homeomorphism between $\tilde{H}$ and $\tilde{H}'$ that induces $\tilde{\sigma}$ on $V(\tilde{H}) \cup E(\tilde{H})$. We may do this edge by edge, as follows. Every edge xy of $\tilde{H}$ is homeomorphic to the edge $\tilde{\sigma}(xy) = \varphi(x)\varphi(y)$ of $\tilde{H}'$, by a homeomorphism mapping x to $\varphi(x)$ and y to $\varphi(y)$. Then the union of all these homeomorphisms, one for every edge of $\tilde{H}$, is indeed a homeomorphism between $\tilde{H}$ and $\tilde{H}'$ – our desired extension of φ to $\tilde{H}$: all we have to check is continuity at the vertices (where the edge homeomorphisms overlap), and this follows at once from our assumption that the two graphs and their individual edges all carry the subspace topology in $\mathbb{R}^3$.

⁴ By the ‘vertices, edges and faces’ of $\tilde{H}$ and $\tilde{H}'$ we mean the images under π^{-1} of the vertices, edges and faces of H and H' (plus $(0, 0, 1)$ in the case of the outer face). Their sets will be denoted by $V(\tilde{H})$, $E(\tilde{H})$, $F(\tilde{H})$ and $V(\tilde{H}')$, $E(\tilde{H}')$, $F(\tilde{H}')$, and incidence is defined as inherited from H and H' .

(4.1.1)
(4.1.4)
(4.2.5)
(4.2.6)

σ

$\tilde{\sigma}$

$\tilde{H}, \tilde{H}'$

In the third step we now extend our homeomorphism $\varphi: \tilde{H} \rightarrow \tilde{H}'$ to all of S^2 . This can be done analogously to the second step, face by face. By Proposition 4.2.6, all face boundaries in $\tilde{H}$ and $\tilde{H}'$ are cycles. Now if f is a face of $\tilde{H}$ and C its boundary, then $\tilde{\sigma}(C) := \bigcup \{ \tilde{\sigma}(e) \mid e \in E(C) \}$ bounds the face $\tilde{\sigma}(f)$ of $\tilde{H}'$. By Theorem 4.1.4, we may therefore extend the homeomorphism $\varphi: C \rightarrow \tilde{\sigma}(C)$ defined so far to a homeomorphism from $C \cup f$ to $\tilde{\sigma}(C) \cup \tilde{\sigma}(f)$. We finally take the union of all these homeomorphisms, one for every face f of $\tilde{H}$, as our desired homeomorphism $\varphi: S^2 \rightarrow S^2$; as before, continuity is easily checked. $\square$

Let us return now to our original goal, the definition of equivalence for planar embeddings. Let us call two planar embeddings ϱ, ϱ' of a graph G *topologically* (respectively, *combinatorially*) *equivalent* if $\varrho' \circ \varrho^{-1}$ is a topological (respectively, combinatorial) isomorphism between $\varrho(G)$ and $\varrho'(G)$. If G is 2-connected, the two definitions coincide by Theorem 4.3.1, and we simply speak of *equivalent* embeddings. Clearly, this is indeed an equivalence relation on the set of planar embeddings of any given graph.

Note that two drawings of G resulting from inequivalent embeddings may well be topologically isomorphic (exercise): for the equivalence of two embeddings we ask not only that some (topological or combinatorial) isomorphism exist between the their images, but that the canonical isomorphism $\varrho' \circ \varrho^{-1}$ be a topological or combinatorial one.

Even in this strong sense, 3-connected graphs have only one embedding up to equivalence:

[12.7.4]

Theorem 4.3.2. (Whitney 1933)

Any two planar embeddings of a 3-connected graph are equivalent.

(4.2.7)

Proof. Let G be a 3-connected graph with planar embeddings $\varrho: G \rightarrow H$ and $\varrho': G \rightarrow H'$. By Theorem 4.3.1 it suffices to show that $\varrho' \circ \varrho^{-1}$ is a graph-theoretical isomorphism, i.e. that $\varrho(C)$ bounds a face of H if and only if $\varrho'(C)$ bounds a face of H' , for every subgraph $C \subseteq G$. This follows at once from Proposition 4.2.7. $\square$

4.4 Planar graphs: Kuratowski's theorem

planar

A graph is called *planar* if it can be embedded in the plane: if it is isomorphic to a plane graph. A planar graph is *maximal*, or *maximally planar*, if it is planar but cannot be extended to a larger planar graph by adding an edge (but no vertex).

Drawings of maximal planar graphs are clearly maximally plane. The converse, however, is not obvious: when we start to draw a planar graph, could it happen that we get stuck half-way with a proper subgraph that is already maximally plane? Our first proposition says that